

Project Description

Sales Performance & Price Elasticity Analysis

This project analyses a retail sales dataset for a single product to evaluate its sales performance, profitability, and customer response to promotional pricing. The objective is to transform raw transactional data into meaningful business insights using SQL, Excel, and Power BI, enabling data-driven decision-making.

The project begins with data preparation and cleaning, followed by the calculation of key business metrics, including Daily Unit Sales Price, Average Unit Sales Price, Gross Profit, Gross Profit Percentage, and Gross Profit per Unit. Promotional periods are identified by analysing changes in unit selling price and quantity sold, allowing for the calculation of the **Price Elasticity of Demand (PED)** to measure how customer demand responds to price changes.

An interactive Power BI dashboard was developed to visualise sales trends, profitability, promotional performance, and key performance indicators (KPIs). The dashboard enables stakeholders to monitor product performance, compare promotional and non-promotional periods, and identify opportunities to optimise pricing strategies.

The project concludes with actionable business recommendations based on the analytical findings, demonstrating how data analytics can support pricing decisions, improve profitability, and enhance overall business performance.

Project Objectives

- Clean and prepare the sales dataset for analysis.
- Calculate key financial and sales performance metrics.
- Analyse gross profit and profitability trends.
- Identify promotional periods using pricing and sales behaviour.
- Calculate the Price Elasticity of Demand (PED) for selected promotions.
- Evaluate whether promotional pricing improves product performance.

- Build an interactive Power BI dashboard for executive reporting.
 - Present business insights and strategic recommendations to support decision-making.
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Tools & Technologies

- **SQL (Databricks SQL)** – Data cleaning, transformation, and metric calculations.
 - **Microsoft Excel** – Data validation, Pivot Tables, and supporting analysis.
 - **Power BI** – Interactive dashboard development and data visualisation.
 - **Data studio** .
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Key Metrics

- Total Sales
 - Quantity Sold
 - Daily Unit Sales Price
 - Average Unit Sales Price
 - Gross Profit
 - Gross Profit Percentage
 - Gross Profit per Unit
 - Promotion Status
 - Price Elasticity of Demand (PED)
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Deliverables

- Cleaned and transformed sales dataset.
- SQL scripts for data preparation and metric calculations.
- Interactive Power BI dashboard.
- Price Elasticity of Demand analysis for selected promotional periods.

- Executive presentation summarising insights and recommendations.
- GitHub repository containing project documentation, SQL scripts, dashboard files, and supporting materials.